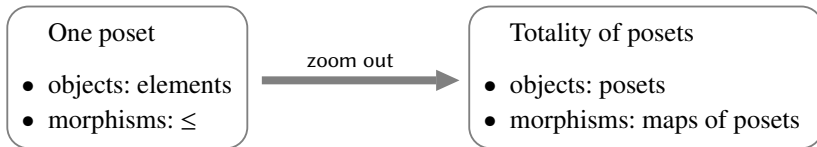


Sets and functions

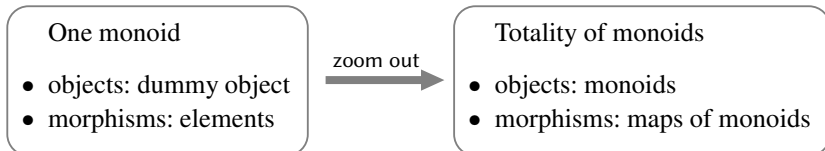
A first example of a large category of mathematical structures. This is one of the fundamental motivating examples of category theory.

So far the examples we have seen have been either tiny drawable examples or examples of individual mathematical structures expressed as a category. However, one of the most powerful aspects of category theory is its ability to zoom out and look at huge worlds of structures rather than just individual structures. Put another way, one of the great insights of category theory is that a totality of mathematical structures of a given type can itself be regarded as a mathematical structure.

That is a lot of words, so here's an example. We saw that a poset is a set of objects equipped with some sort of ordering on it. We expressed that as a category, where the objects are the elements of the given poset. But we could now look at the collection of *all possible posets* and regard that as a category. The objects of this category are now *the posets themselves* and the arrows will be some form of map between posets. Here's a schematic diagram.



We could do this for monoids as well:



We could also look for a category of groups and maps between groups, or spaces and maps between spaces.

The starting point for all of these structures is an underlying set. We then say

that the set is special in some way, by having an ordering, or by having a binary operation, or by having a notion of closeness. The starting point for the *maps* between structures is then the concept of a function, the basic notion of a map between sets. We then find a good notion of map between special structures by looking at maps that “preserve” the special properties in question. We’ll look at that, but first we need to think about functions.

12.1 Functions

The category of sets and functions is one of the key motivating examples in category theory. Sets and functions are an essential starting point of the whole of mathematics, and the techniques mathematicians have built up for dealing with them are things we can then try and do in more complicated and nuanced mathematical worlds.

In a way, numbers are simplified versions of sets. You might think that numbers are the fundamental concepts of mathematics, and in a sense that is true both historically and in terms of the order in which we tend to learn and understand math as children. However, when mathematicians tried to make the idea of numbers very precise, they found they had to deal with sets first, and then build numbers out of particular sets. When mathematicians became really serious about developing mathematics completely rigorously, they started with sets. Set theory was developed as a *foundation* of mathematics.

Category theory is sometimes considered to be a competing foundation. I don’t really like competitions and don’t see why everything has to be made into a competition, so I don’t want to start a set theory vs category theory flame war (there are plenty of those on the internet). But I’ll just say this. Set theory starts with the concept of *membership* as a basic concept, and builds up from the idea of elements being members of sets. By contrast, category theory builds up from the notion of *relationships* which is why our basic concepts in category theory are elements (objects) and relationships (arrows) between them, rather than membership.

Ideologically I like to think of this as the difference between looking at people’s intrinsic characteristics and looking at how they behave in different contexts; I wrote about that at length in my book $X + Y$. But a more mathematical ideology to consider is that category theory takes *sets and functions* as the basic starting point, rather than sets and membership.

The two approaches are in some sense equivalent, in that you can build the concept of a function from the concept of set membership, and conversely you

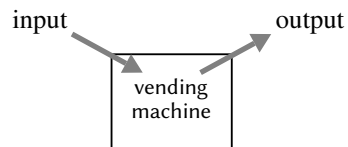
can build the concept of set membership from the concept of a function. It's really a matter of preference which concept you take to be more fundamental.

Many people, including me, find that the categorical approach more closely matches my actual mathematical practice. Gathering together mathematical structures and thinking about maps between them seems to happen all over pure math, and so taking functions as a starting point seems more illuminating to me. I think this is part of why the language and formalism of category theory have turned out to be so fruitful across many branches of math.

Anyway, this means we'd better develop an understanding of functions. Functions are quite misunderstood. The trouble is that, like with so many parts of math, the way they're typically treated in school math is not really like the way they're treated in research math. For that reason this chapter might actually be easier if you've never encountered functions before; if you've encountered them before in high school math classes then you might have a bit of un-remembering to do (unremembering?) in order to get your head around functions in the way that category theorists think of them.

In school you might have met functions as things like $f(x) = x^2$ or $\sin$, $\cos$, $\tan$ or e^x . Those are all examples of functions but they are not really expressed categorically, as this doesn't tell us what sets we are relating to each other. In a categorical approach, a function is a relationship between sets. It goes from one set to another. We can think of those sets as set of inputs and a set of potential outputs. It's like a machine, where you can feed in any of the inputs and one of the outputs will pop out. The inputs and outputs could be any sets: they don't have to be numbers, although functions that deal with numbers are a big part of math and those tend to be the kinds of functions you meet in school. Moreover, functions don't have to be given by any kind of formula although, again, those are the kind you tend to meet in school.

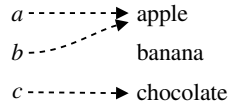
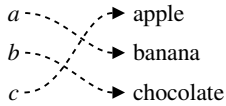
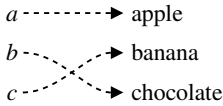
A function could be something like a vending machine.



Perhaps the machine has three buttons, labeled a, b, c , and you can buy an apple, a banana, or chocolate. This chart tells us what each button produces.

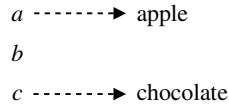
a	----->	apple
b	----->	banana
c	----->	chocolate

This is the most “sensible” way to set up this particular machine, but it doesn't have to be like that. It could instead be set up in any of the following ways, for example. The third might happen if there are no bananas today.

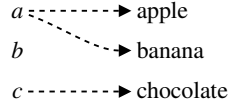


It's important to note that in the last case the banana still counts as an output even though there's no way of achieving it, so it might be better to think of the outputs as *potential outputs*.

It's also important to note that all the inputs have to produce something. If we ran out of bananas and just left the "b" slot empty then it would not count as a function any more.



Also we can't put two different items in the same compartment: when we press a button we have to know exactly what we are going to get out.



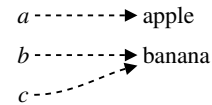
Those other situations do exist, they're just not counted as functions. So functions have an asymmetry between the role of the inputs and the outputs:

- Every input must produce an output. However, there could be outputs that are not produced by inputs.
- We can't have one input producing two different outputs. However, we can have one output being produced by two different inputs.

Functions with less asymmetry in the role of input and output are interestingly well-behaved; we'll look at them later when we think about special properties of functions.

If there is a different number of inputs from outputs we are destined to have either some outputs left out, or some outputs hit by more than one input.

For example, if chocolate is completely removed from the vending machine options, we'll only have two possible outputs. Any function will need to have at least two inputs producing the same output, like here.



Things To Think About

T 12.1 How many different functions are possible between these three inputs and two outputs above? You could try drawing them all or you could think through the problem logically. What if there were p inputs and k outputs?

To see how many functions there are we could take each input in turn. The first input has two possible choices for the output. The second input also does, and so does the third. The choice of output for a does not affect any of the

other possible outputs, so these choices are all independent, which means we can just multiply the number of choices to get $2 \times 2 \times 2 = 8$.

We could make a list of all the possible functions in a table as follows. Here, each column represents one function: the top entry is the output for a , the middle entry is the output for b , and the bottom entry is the output for c .

a	apple	apple	apple	apple	banana	banana	banana	banana
b	apple	apple	banana	banana	apple	apple	banana	banana
c	apple	banana	apple	banana	apple	banana	apple	banana

If we generalize to p inputs and k outputs we see that each input has a choice of k outputs, so the total number of possible functions is: $\underbrace{k \times \cdots \times k}_{p \text{ times}} = k^p$.

Things To Think About

T 12.2 What does this formula say if the set of inputs is the empty set? What if the set of outputs is the empty set? Is this weird? Can you interpret it?

The empty set (which we write as $\emptyset$) is a set containing 0 elements. So if we have 0 inputs then $p = 0$ and the formula tells us that the total number of possible functions is k^0 which is 1 no matter what k is. This seems to be telling us that no matter what set of outputs B we pick there is one function $\emptyset \rightarrow B$.

How can that be? If there are no inputs then how can there be a function? This is an example where it may be necessary to take the definition formally rather than informally. The formal definition says something like “ f produces, for every object in A , an object of B ”. The key is in the phrase “for every object in A ”. If A has no objects then we are already done: whatever comes after that clause, we have already done it for every object in A , because there aren’t any. This is called having a condition that is “vacuously satisfied”. For example all the elephants in my house are purple. This is true because there are no elephants in my house. The “vacuous” function from the empty set to any set is something we’ll come back to when we talk about initial objects. It’s not exactly interesting in its own right, but it is an interesting and indeed important function to have around in the context (category) of sets and functions.

On the other hand if we have 0 outputs then we put $k = 0$ in the formula and the total number of possible functions is 0^p , which is 0 unless there are also zero inputs, in which case the answer is 1. That is not surprising — a function has to produce an output for every input, so if there are no outputs we’ll be really stuck unless there are also no inputs.

The above discussion has, I hope, made it clear that we really should define functions a bit more formally. Here is a more formal definition. You might find the wording strange, especially the grammar involving the word “assigns”,

which is not really the way we use that verb in normal English. I think we tend to word it this way round in math so that the inputs come before the outputs in the sentence.[†]

Definition 12.1 Let A and B be any sets. A *function* $A \xrightarrow{f} B$ assigns to every element $a \in A$ an element $f(a) \in B$.

In very formal set theory the definition happens by means of “ordered pairs”: we construct a new set from A and B consisting of pairs of elements, one from A and one from B , representing the input–output pairs that define f . We will not need to go to that level of formality here.

Notation and terminology

In category theory thinking, a function *always* comes with the information of what its sets of inputs and outputs are. Functions are going to be the arrows of a category, as you have probably guessed. So we might call A and B the source and target of the function f , or the domain and codomain. They are sometimes called the domain and range, but I don’t like “range” because it is used ambiguously in school math.

We might write this function as $f: A \rightarrow B$ or as $A \xrightarrow{f} B$. That is at the level of sets. If we start thinking about the action of f on elements inside the sets we write things like $a \in A, f(a) \in B$. If $f(a) = b$ we might say b is the *image* of a under f , and we might write $f: a \mapsto b$. The vertical bar on the arrow is to indicate that we are now talking about the action of the function on elements, not about its source and target sets.

We might sum up the whole situation as shown on the right; note the two different types of arrow meaning different things with respect to the same function f .

$$\begin{array}{ccc} f : A & \longrightarrow & B \\ a & \longmapsto & b \end{array}$$

In school math, functions are often defined without specifying what their source and target sets are, and then you are somehow supposed to infer what their domain and “range” are based on where it is possible to define them. That makes me very uncomfortable categorically (as well as linguistically).

Sometimes functions can be defined by a formula, but not always, and sometimes their inputs and outputs aren’t actually numbers.

[†] Perhaps a more natural way to phrase this in normal English would be something like “A function assigns an element of B to each element of A .”

Examples Here are some examples from life and then some examples from math. We will think more about these examples in Chapter 15 when we think about particular properties of functions. Each one is a function $A \xrightarrow{f} B$.

- | | |
|---|---|
| 1. $A =$ people in Chicago
$B =$ countries in the world
$f(a) =$ where a was born. | 3. $A = \{1, 2, 3\}$
$B = \{1, 2, 3, 4, 5\}$
$f(a) = a + 1$ |
| 2. $A =$ people in New York City
$B =$ the set of pairs {name, date}
$f(a) =$ the name and date of birth of the person. | 4. $A = \mathbb{Z}$
$B = \mathbb{Z}$
$f(a) = a + 1$ |

We are of course going to assemble sets and functions into a category. The objects will be all possible sets and the arrows $A \rightarrow B$ will be all the possible functions $A \rightarrow B$. An aside is probably called for here on what a set is. I don't intend to get into the foundations of set theory and it's not *really* necessary in order to think about the category of sets. I think we just need to agree that we have a notion of set such that a set is entirely determined by its elements, and such that Russell's paradox doesn't happen, as we mentioned in Chapter 8. (Again, see Appendix C for more details.)

Note that a set is completely determined by its elements and a function is completely determined by what it does to elements. So it doesn't matter how it produces the outputs — it could be by a formula, by a machine, by looking at a list — if two functions take the same inputs and produce the same outputs for each one then they count as the same function.[†] (Note that in category theory they must also have the same source and target sets before we can even ask if they are the same function.)

So for example we could define two functions $\mathbb{R}_+ \rightarrow \mathbb{R}$ like this:

$$\bullet f(x) = \frac{x}{x} \qquad \bullet g(x) = 1.$$

Note that $\mathbb{R}_+$ means all the *positive* real numbers, so we don't have to worry about what $\frac{0}{0}$ is. Anyway for all $x \in \mathbb{R}_+$, we have $f(x) = g(x) = 1$ and this means that $f = g$ as functions, even though they have different formulae.

Definition 12.2 Let f, g be functions $A \rightarrow B$. We say $f = g$ as functions whenever $\boxed{\forall a \in A \ f(a) = g(a)}$.

We now have the data for our category of sets and functions, that is the objects (sets) and the morphisms (functions); next we need to sort out the structure and the properties.

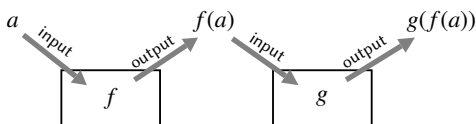
[†] This is the essence of the definition of a function in set theory by input–output pairs.

12.2 Structure: identities and composition

Given any set A there is an identity function $A \rightarrow A$ which just sends every element of A to itself. We can call it 1_A and define it by $\boxed{\forall a \in A \ 1_A(a) = a.}$

For composition we need to think about functions $A \xrightarrow{f} B \xrightarrow{g} C$. We can think about defining the composite function as if we are joining two machines together.

Here's a picture showing that we can put an input into f , and then whatever the output is, we immediately feed that into g and see what the output is.



Here's the more mathematically rigorous presentation.

$$\begin{array}{ccccc}
 A & \xrightarrow{f} & B & \xrightarrow{g} & C \\
 a & \mapsto & f(a) & \mapsto & g(f(a))
 \end{array}$$

So we define the composite function $g \circ f$ by: $\boxed{(g \circ f)(a) = g(f(a)).}$

Notation for composites

This finally explains why the composition notation for arrows in a category looks “backwards”: when we annotate functions, we put the variable (input) on the right so that the function can act on it from the left. But that means that a subsequent function will need to act on the left of that, which means that composites build up to the left even if the arrows are heading towards the right. Some people try and make this consistent by drawing arrows pointing to the left, or by writing functions acting on the right. I prefer to just give up on the idea of consistency. Incidentally I think that is the most widespread practice among mathematicians. Drawing arrows pointing left is quite unusual; writing functions acting on the right is less unusual but still not mainstream (although it may be more common in some countries or in some subjects).

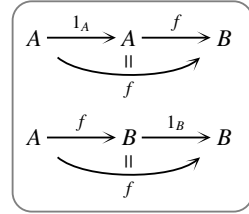
Things To Think About

T 12.3 To show that sets and functions form a category we need to check that the identity really behaves like an identity, both on the left and on the right, and that composition is associative. We'll do that next, but you might want to see if you can check for yourself first.

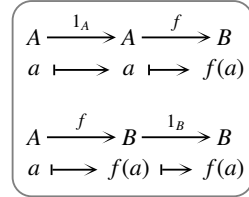
12.3 Properties: unit and associativity laws

For properties we have to check unit laws and associativity.

For the unit laws we need to check that the identity function behaves like the identity on each side, that is, given a function $A \xrightarrow{f} B$ we need to check these two equations hold.



We can “chase” elements through each diagram as shown here. This means we consider an element in the set at the start of the diagram and follow it through arrow by arrow to the end to see where it arrives.



In both cases the element a arrives at $f(a)$, showing that in each case the composite is indeed the same as f . In more traditional algebra we might write it like this: for any $a \in A$

$$\begin{aligned} (f \circ 1_A)(a) &= f(1_A(a)) \text{ by definition of } \circ \\ &= f(a) \text{ by definition of } 1_A \end{aligned}$$

as required. (And similarly on the other side.)

To check that composition of functions is associative we need to consider three composable functions $A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D$ and show that

$$(h \circ g) \circ f = h \circ (g \circ f)$$

$$\text{that is, } \forall a \in A: ((h \circ g) \circ f)(a) = (h \circ (g \circ f))(a).$$

This is really just a case of carefully writing out the definitions and getting the parentheses in the right places. I recommend you try it yourself to see if you really understand; in any case you might find that less painful than trying to read through this printed version, but here goes anyway. For any $a \in A$:

The left-hand side gives us

$$\begin{aligned} ((h \circ g) \circ f)(a) &= (h \circ g)(f(a)) \\ &= h(g(f(a))) \end{aligned}$$

The right-hand side gives us

$$\begin{aligned} (h \circ (g \circ f))(a) &= h((g \circ f)(a)) \\ &= h(g(f(a))) \end{aligned}$$

So the two sides are equal as required.

12.4 The category of sets and functions

We now have all we need to construct the very important category of sets and functions. This category is typically written as **Set** and is given by:

- objects: all sets
- morphisms: a morphism $A \longrightarrow B$ is a function $A \longrightarrow B$
- identities: given a set A , the identity is the identity function 1_A
- composition: composition of morphisms is composition of functions.

This category is not only the starting point for building many other categories, it is also the prototype for a “well-behaved” category and the motivation for many of the structures we seek to express in a general category. The idea of **Set** as a starting point is that any mathematical structure that starts with an underlying set together with some other stuff can then be built from the category **Set** by looking at “sets with stuff” and “functions interacting well with that stuff”. We’ll look at this next.

The idea behind **Set** as a prototype is that this category is the basic world in which much of mathematics is done. So we know it is a good world in which we can do many things we like (if we like doing math at all). Then if we move to work in a different world we might want to be able to do some of those same things, so we work out exactly what aspect of the category **Set** enabled us to do those things, and then look for those features in the new category we want to work in. In order to do that we have to express those features just using the objects and arrows of the category **Set**, not any particular details about sets themselves. This is the idea behind expressing things categorically, rather than at the level of elements (of sets), and it will come up a lot when we start working with category theory.